Electric plug-in evaporators: HMTc Standard

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CORRESPONDENCE

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heavy metals, food safety, dietary exposure

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for electric plug-in evaporators in Home Air and Inhalation-Adjacent Products, evaluated on as sold, micrograms of element per kilogram of the emitting material (the fragrance concentrate, wax, gel, botanical mass or refill charge that is aerosolised, evaporated or combusted), with component-level screening of the article's non-emitting parts; a content screen on the article and its refill, never an emission rate, a per-actuation dose or an indoor-air concentration. 4 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only initial setting: the strictest applicable sovereign maximum for this exact product form in its native basis; disclosed read-across where the exact cell is unregulated; control where neither a sovereign number nor a supportable near-neighbour analogue exists; never occurrence-derived)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	500	human-grade read-across (US-CO)
Cadmium (Cd)	Tier 1	200	human-grade read-across (US-CO)

Total arsenic (tAs)	Tier 2	200	human-grade read-across (US-CO)
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	100	human-grade read-across (US-CO)
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	600	screening ceiling (US-MD)
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	US-CO — human-food analog
Cadmium (Cd)	200	US-CO — human-food analog
Total arsenic (tAs)	200	US-CO — human-food analog
Inorganic arsenic (iAs)	control	No government sets an inorganic-arsenic content maximum for any inhaled consumer product or any home-air product. Every route-scoped inhaled-product instrument regulates arsenic as the unspeciated total: Colorado, California, Maryland, Massachusetts, Nevada and Israel all publish total arsenic and none of them speciates, so there is no inorganic-arsenic cell anywhere to read across from. The article and component instruments checked were EU RoHS Directive 2011/65/EU Annex II (lead, cadmium, total mercury and hexavalent chromium in the homogeneous materials of electrical and electronic equipment), the EU Batteries Regulation Annex I (mercury, cadmium and lead in batteries), REACH Annex XVII entry 23 (cadmium in synthetic polymers, paint and painted articles), REACH Annex XVII entry 20 (organotin expressed as tin in mixtures

Mercury, total (tHg)	100	US-CO — human-food analog
Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic sediment and biomagnifies through aquatic food webs. This row's emitting material is a manufactured non-marine formulation (the refill fragrance oil charge) with no methylation pathway, no aquatic trophic transfer and no marine-derived or wild-harvested animal input, and methylmercury is in any event a component of total mercury, so MeHg is less than or equal to tHg by identity and is bounded by and controlled through the published total-mercury ceiling of 100 ug/kg. Methylmercury is additionally thermally labile and decomposes well below the temperatures reached in any heated or combustion form in this category, so no thermal pathway creates it either. No credible sovereign government sets a methylmercury limit for any air-care product, and no routed occurrence source reports methylmercury in one, which is the speciation-aware condition the not-material integrity gate applies. No separate methylmercury ceiling is published because none would bind below the total-mercury ceiling.

Nickel (Ni)	control	No government sets a nickel content maximum for any inhaled consumer product or any home-air product. Every route-scoped inhaled-product instrument on the register omits nickel entirely, including Maryland's inhalation-route table, which publishes lead, arsenic, mercury, cadmium and total chromium but no nickel. The nearest nickel instruments were checked and each fails on basis or on route: the EU Toy Safety Directive nickel values are element MIGRATION limits from toy material under EN 71-3, not content limits; REACH Annex XVII entry 27 governs nickel RELEASE RATE from skin-piercing and prolonged--skin-contact articles; Korea MFDS 2025-63 and 2024-9 set nickel content limits for cosmetics on a dermal basis; and EU Regulation (EU) 2024/1987 sets dietary nickel maximum levels on an ingestion basis. No conversion between a migration limit, a release rate, a dermal cosmetic limit and a total-content limit for an inhaled product is defensible, and none is performed. A not-material determination is rejected: Nickel is material here: plug pins, spring contacts and heater terminations are routinely nickel-plated, they run warm for weeks at a time, and the heated wick sits in the emission path immediately downstream of them. The cell is a control. No occupational exposure limit, indoor-air guideline or ambient-air standard was used, and none was converted into a product ceiling: an air concentration is not a product content maximum and HMTc does not manufacture one from the other.

Tin (Sn)	control	No government sets a total-tin content maximum for any inhaled consumer product or any home-air product. The tin instruments on the register were checked and each fails on basis or on scope: the Codex, EU, Hong Kong and Japan canned-food tin maximum levels govern tin dissolved from tinplate INTO A FOOD on a dietary as-consumed basis; the EU Toy Safety Directive tin values are EN 71-3 migration limits from toy material; and REACH Annex XVII entry 20 restricts ORGANOTIN SPECIES expressed as tin in mixtures and articles supplied to the general public, which does directly cover the articles in this row but is a species restriction at 0.1 percent by weight rather than a total-tin content ceiling, and adopting 1,000,000 ug/kg as this row's tin limit would be to publish a number that protects nobody. It is recorded as the companion instrument and enforced as a precondition instead. A not-material determination is rejected: Tin is material here: every solder joint on the printed circuit assembly is a tin alloy, plug-pin and terminal plating is frequently tin, and organotin catalyst residues occur in silicone gaskets and seals. The cell is a control. No occupational exposure limit, indoor-air guideline or ambient-air standard was used, and none was converted into a product ceiling: an air concentration is not a product content maximum and HMTc does not manufacture one from the other.

Aluminium (Al)	control	No government sets an aluminium content maximum for any inhaled consumer product or any home-air product. The aluminium instruments on the register were checked and each fails on basis or on route: the EU Toy Safety Directive aluminium values are EN 71-3 element migration limits from toy material, and the dietary and food-additive aluminium provisions are calibrated to oral intake and oral absorption. Aluminium is also the panel analyte for which route-transfer is least defensible in principle, because a dietary aluminium limit is written against gastrointestinal absorption of well under one percent, which does not describe deposition in the respiratory tract. HMTc states that asymmetry and does NOT convert it into a number: no exposure model, no percentile
Total chromium (Cr)	600	government maximum level
Hexavalent chromium (Cr-VI)	control	

4 | Cite this standard

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References

[1] US-CO — human-food maximum levels
Binds Pb, Cd, tAs, tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/electric-plug--in-evaporators. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/electric-plug--in-evaporators.